A Chern Number for Gauge Fields on $\mathbb{R}^4$

by Roger Schlafly

1. Introduction

Let G be a compact Lie group and let $\rho: G \to U(N)$ be a representation. Let F be the curvature of some connection on some principal G-bundle over S^4. The Chern number of the associated vector bundle is

$$c_2 = \frac{1}{8\pi^2} \int_{S^4} \mathrm{Tr}\, \rho(F)^2 .$$

According to the theory of characteristic classes, this quantity is always an integer.

Now consider the curvature F of a connection on a principal G-bundle over $\mathbb{R}^4$. In this paper, we prove

Theorem 1.1. If

$$|F| \leq \frac{c}{r^2 \log r}, \quad r \geq 2$$

for some constant c, then

$$\frac{1}{8\pi^2} \int_{\mathbb{R}^4} \mathrm{Tr}\, \rho(F)^2$$

is an integer.

This formula is of interest in quantum field theory because the curvature F is the same as a Yang-Mills (gauge) field over Euclidean space. For discussions of related matters, see [1], [2] and [11].

If G is abelian or $N = 1$, the integer obtained in Theorem 1.1 must be zero. If $G = SU(N)$ $(N > 1)$, any integer value is possible.

It is not known whether the hypothesis of Theorem 1.1 can be relaxed to the energy $\int_{\mathbb{R}^4} |F|^2$ being finite, although this is suggested by [2], [6], and [9]. However, it is shown in section 6 that $\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \rho(F)^2$ may not be an integer if we assume only that $|F| \leq c/r^2$. This example is equivalent to the one given in [9], and has infinite energy.

Theorem 1.1 is proved in sections 2, 3, 4, and 5. Section 3 gives a holonomy formula similar to the one in [10] and may be read independently of the rest of the paper. Section 7 shows how to handle dimensions other than four.

This paper was motivated by a question raised by I. M. Singer. I would like to thank him for many interesting conversations.

2. Outline of the Proof

For the proof of Theorem 1.1 it suffices to consider the associated $U(N)$ bundle. Hence we may, without loss of generality, take $G = U(N)$ and suppress mention of ρ.

Relative to some trivialization of the bundle, the connection A is a one-form on $\mathbb{R}^4$ with values in $u(N)$. The curvature

$$F = dA + A^2$$

is a two-form on $\mathbb{R}^4$ with values in $u(N)$. The Chern-Weil formalism [4, p. 114] tells us that

$$\operatorname{Tr} F^2 = d \operatorname{Tr}(AF - \tfrac{1}{3}A^3) .$$

Let S^3_r be the sphere of radius r in $\mathbb{R}^4$. If $|F| \leq \frac{c}{r^2 \log r}$. Then $\operatorname{Tr} F^2$ is integrable on $\mathbb{R}^4$. By Stokes' theorem,

$$\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 = \lim_{r\to\infty} \frac{1}{8\pi^2}\int_{S^3_r} \operatorname{Tr}(AF - \tfrac{1}{3}A^3) .$$

Given $\epsilon > 0$ and r sufficiently large, we will show that there exists a smooth map $\tilde{T}: S^3_r \to U(N)$ such that

$$\left|\frac{1}{8\pi^2}\int_{S^3_r} \operatorname{Tr}[AF - \tfrac{1}{3}A^3 - \tfrac{1}{3}(\tilde{T}^{-1}d\tilde{T})^3]\right| < \frac{\epsilon}{2}.$$

If we choose r large enough such that

$$\left|\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 - \frac{1}{8\pi^2}\int_{S^3_r} \operatorname{Tr}(AF - \tfrac{1}{3}A^3)\right| < \frac{\epsilon}{2},$$

then it will follow that

$$\left|\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 - \frac{1}{24\pi^2}\int_{S^3_r} \operatorname{Tr}(\tilde{T}^{-1}d\tilde{T})^3\right| < \epsilon.$$

We now recall a special case of Bott's work on periodicity in K-theory [3].

Theorem 2.1. For $N \geq 2$ we have an isomorphism $\pi_3 U(N) \cong \mathbb{Z}$ given by assigning to a smooth map $\tilde{T}: S^3 \to U(N)$ the integer

$$\frac{1}{24\pi^2}\int_{S^3} \operatorname{Tr}(\tilde{T}^{-1}d\tilde{T})^3 .$$

Thus for each $\epsilon > 0$ there exists an integer n such that

$$\left|\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 - n\right| < \epsilon.$$

Hence $\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \mathrm{Tr}\, F^2$ must be an integer.

3. The Holonomy Propagator

Let M be the sector in $\mathbb{R}^2$ described by polar coordinates (r,θ) with $0 \leq r \leq r_0$ and $0 \leq \theta \leq \theta_0$. Suppose we have a connection A on the trivial principal G-bundle. For this paper we can assume that G is $U(N)$, although Theorems 3.1 and 3.2 hold for any Lie group. A is a one-form on M with values in $u(N)$. The curvature is $F = dA + A^2$.

For $x \in M$ let $c_x\colon [0,1] \to M$ be the curve

$$c_x(t) = tx\ , \quad 0 \leq t \leq 1\ .$$

A section $u\colon M \to \mathbb{C}^N$ of the associated vector bundle is parallel along c_x iff it satisfies

$$c_x^*(du + Au) = 0\ .$$

The fundamental solution to this differential equation is the path ordered exponential

$$P_{exp} - \int_{c_x} A\ .$$

P_{exp} should be $Pexp$

If u is parallel along c_x then

$$u(x) = P_{exp} - \int_{c_x} A\cdot u(0)\ .$$

$P_{exp} - \int_{c_x} A$ is called the holonomy from 0 to x along c_x. It may also be defined as a product integral [8, p. 15].

Theorem 3.1. If

$$T(x) = P_{exp} - \int_{c_x} A$$

then

$$A(x) = -dT\cdot T^{-1} + T(\int_{c_x} T^{-1}FT)T^{-1} .$$

Proof. Plugging in the vector $\partial/\partial r$, we get

$$A(\frac{\partial}{\partial r}) = -\frac{\partial T}{\partial r}\cdot T^{-1}$$

which follows from the definition of T. The vector $\partial/\partial\theta$ gives

$$A_2 = -\frac{\partial T}{\partial \theta}\cdot T^{-1} + T(\int_0^r T^{-1}F_0T\ dr)T^{-1}$$

where

$$A = A_1 dr + A_2 d\theta$$
$$F = F_0\ dr\wedge d\theta .$$

To prove this, we first observe that

$$-\frac{\partial A_1}{\partial\theta} = \frac{\partial}{\partial\theta}(\frac{\partial T}{\partial r}\cdot T^{-1}) = \frac{\partial^2 T}{\partial\theta\partial r}\cdot T^{-1} - \frac{\partial T}{\partial r}T^{-1}\frac{\partial T}{\partial\theta}T^{-1} .$$

Hence

$$\begin{aligned}\frac{\partial}{\partial r}[T^{-1}(A_2 + \frac{\partial T}{\partial\theta}T^{-1})T] &= -T^{-1}\frac{\partial T}{\partial r}T^{-1}A_2T + T^{-1}\frac{\partial A_2}{\partial r}T + T^{-1}A_2\frac{\partial T}{\partial r}\\ &\quad - T^{-1}\frac{\partial T}{\partial r}T^{-1}\frac{\partial T}{\partial\theta} + T^{-1}\frac{\partial^2 T}{\partial r\partial\theta}\\ &= T^{-1}A_1A_2T + T^{-1}\frac{\partial A_2}{\partial r}T - T^{-1}A_2A_1T - T^{-1}\frac{\partial A_1}{\partial\theta}T\\ &= T^{-1}F_0T .\end{aligned}$$

Continuity of A and $dT\cdot T^{-1}$ at 0 require that

$$\lim_{r\to 0} A_2 = 0$$
$$\lim_{r\to 0} \frac{\partial T}{\partial\theta}\cdot T^{-1} = 0$$

so

$$T^{-1}(A_2 + \frac{\partial T}{\partial \theta} T^{-1})T = \int_0^r T^{-1} F_0 T \, dr$$

which gives the formula we wanted. //

Theorem 3.2. The holonomy from 0 to 0 along ∂M is given by the θ-ordered exponential

$$P_{exp} - \int_M T^{-1} F T \ .$$

Proof. Let

$$V(r,\theta) = P_{exp} \int_0^\theta \int_0^r T^{-1} F_0 T \, dr d\theta$$

with the ordering over θ, so

$$V(r_0,\theta_0) = P_{exp} - \int_M T^{-1} F T \ .$$

By Theorem 3.1,

$$V = P_{exp} - \int_0^{\theta_0} (T^{-1} A_2 T + T^{-1} \frac{\partial T}{\partial \theta}) d\theta \ .$$

We compute

$$\begin{aligned} \frac{\partial}{\partial \theta}(TV)(TV)^{-1} &= T \frac{\partial V}{\partial \theta} V^{-1} T^{-1} + \frac{\partial T}{\partial \theta} T^{-1} \\ &= -T(T^{-1} A_2 T + T^{-1} \frac{\partial T}{\partial \theta}) T^{-1} + \frac{\partial T}{\partial \theta} T^{-1} \\ &= -A_2 \end{aligned}$$

so by the uniqueness of solutions to this ordinary differential equation, there must be some $W: M \longrightarrow G$ independent of θ such that

$$TV = P_{exp} - \int_0^\theta A_2 d\theta \cdot W \ .$$

Writing $T = T(r,\theta)$ and letting $\theta = 0$ we find that $W = T(r,0)$ so

$$T(r,\theta)V(r,\theta) = P_{exp} - \int_0^\theta A_2 d\theta \cdot T(r,0) .$$

Setting $r = r_0$ and $\theta = \theta_0$ gives

$$V(r_0,\theta_0) = T(r_0,\theta_0)^{-1} P_{exp} - \int_0^{\theta_0} A_2 d\theta \cdot T(r_0,0)$$

which is precisely the holonomy from 0 to 0 along ∂M. //

Proposition 3.3. For any one-form B and any curve c,

(i) $|P_{exp}\int_c B| \leq \exp \int_c |B|$

(ii) $|1 - P_{exp}\int_c B| \leq \int_c |B| \exp \int_c |B|$

Proof. Straightforward. //

Corollary 3.4. If H is the holonomy along ∂M then

$$|1-H| \leq \int_M |F| \cdot \exp \int_M |F| .$$

4. Estimates in the Radial Gauge

We now choose a gauge in which there is no holonomy in the radial direction. This means that $A(\partial/\partial r) = 0$ where r is the distance to the origin in $\mathbb{R}^4$. Define $G: \mathbb{R}^4 \to U(N)$ by

$$G(x) = P_{exp} - \int_{c_x} A$$

where

$$c_x(t) = tx , \quad 0 \leq t \leq 1 .$$

Proposition 4.1. (i) Replacing A by $G^{-1}AG + G^{-1}dG$ (and F by $G^{-1}FG$) gives us a radial gauge.

(ii) In this gauge,

$$A(x) = \int_{c_x} F \ .$$

(iii) If c is a constant such that

$$|F| \leq \frac{c}{r^2 \log r}$$

for $r \geq 2$ and

$$|A| \leq \frac{c}{e} \log \log 3$$

for $r = e = 2.718...$, then for $r \geq 3$ we have

$$|A| \leq 2c \frac{\log \log r}{r} \ .$$

Proof. (i) and (ii) follow from Theorem 3.1. For (iii), write $F = F_0 \, dr \wedge d\theta$ so

$$A = \left(\int_0^r F_0 dr \right) d\theta \ .$$

Since $|d\theta| = \frac{1}{r}$ and $|F| = |F_0| \cdot \frac{1}{r}$, we have

$$\begin{aligned} |A| &\leq \left| \int_0^e F_0 dr \right| \cdot |d\theta| + \left| \int_e^r F_0 dr \right| \cdot |d\theta| \\ &\leq \frac{c}{r} \log \log 3 + \frac{1}{r} \int_e^r \frac{c}{r^2 \log r} \, r dr \\ &= \frac{c}{r} \log \log 3 + \frac{c}{r} \cdot \log \log r \\ &\leq 2c \frac{\log \log r}{r} \ . \end{aligned}$$

//

Proposition 4.2. For $r \geq 3$, there exists a constant K_1 such that

$$\left| \iint_{S_r^3} \operatorname{Tr} AF \right| \leq K_1 N c^2 \cdot \frac{\log \log r}{\log r} .$$

Proof. The volume of S_r^3 is $2\pi r^3$, so

$$\begin{aligned} \left| \iint_{S_r^3} \operatorname{Tr} AF \right| &\leq \int_{S_r^3} |\operatorname{Tr} AF| \\ &\leq N \int_{S_r^3} |A| \cdot |F| \\ &\leq N \cdot 2\pi r^3 \cdot 2c \, \frac{\log \log r}{r} \cdot \frac{c}{r^2 \log r} \\ &= 4\pi N c^2 \cdot \frac{\log \log r}{\log r} . \end{aligned}$$

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Proposition 4.3. Suppose $r \geq 3$. If

$$|A-B| \leq \frac{2\pi c}{r \log r} ,$$

then there exists a constant K_2 such that

$$\left| \iint_{S_r^3} \operatorname{Tr} A^3 - \int_{S_r^3} \operatorname{Tr} B^3 \right| \leq K_2 N c^2 \cdot \frac{(\log \log r)^2}{\log r} .$$

Proof. We have

$$\begin{aligned} |B| &\leq |B-A| + |A| \\ &\leq \frac{2\pi c}{r \log r} + 2c \, \frac{\log \log r}{r} \\ &\leq 4\pi c \, \frac{\log \log r}{r} \end{aligned}$$

and

$$A^3 - B^3 = A^2(A-B) + A(A-B)B + (A-B)B^2$$

so

$$|A^3-B^3| \le \frac{2\pi c}{r^2 \log r} \cdot 3\left(\frac{4c}{r}\frac{\log\log r}{r}\right)^2$$
$$= \frac{96c^3}{r^3 \log r}(\log\log r)^2 .$$

Thus

$$\left|\int_{S_r^3} \mathrm{Tr}\, A^3 - \int_{S_r^3} \mathrm{Tr}\, B^3\right| \le N \int_{S_r^3} |A^3-B^3|$$
$$\le N \cdot 2\pi r^3 \cdot \frac{96c^3}{r^3 \log r}(\log\log r)^2$$
$$\le 300Nc^2 \cdot \frac{(\log\log r)^2}{\log r} . \qquad //$$

5. The Holonomy at Infinity

In this section, we restrict attention to S_r^3 for sufficiently large r. $C_1, C_2, \ldots, C_{13}$ will be constants independent of r.

Fix some "north pole" $p \in S_r^3$, and some great circle Γ from p to $-p$. If $x \in S_r^3$ is not on the "equator" S_r^2, let c_x be the shorter great arc from $\pm p$ to x. (There is a unique great circle in S_r^3 passing through $\pm p$ and x if $x \neq \pm p$.) For x in the "northern hemisphere," let $T(x)$ be the holonomy along c_x. For x in the "southern hemisphere", let $T(x)$ be the holonomy along Γ concatenated with c_x. T is discontinuous on the equator S_r^2.

Lemma 5.1. There exists a constant c_1 such that if T_1, T_2 are the two limiting values of T at a point x in S_r^2, then

$$|T_1-T_2| \le \frac{c_1}{\log r} .$$

Proof. Choose a surface S in S^3_r having area $\leq 2\pi r^2$ and boundary the union of Γ with the great semicircle from p to $-p$ through x. Then

$$\int_S |F| \leq 2\pi r^2 \cdot \frac{c}{r^2 \log r} = \frac{2\pi c}{\log r} .$$

The holonomy around ∂S is $T_1 T_2^{-1}$, so by Corollary 3.4,

$$|T_1 - T_2| = |1 - T_1 T_2^{-1}| \leq \int_S |F| \exp \int_S |F| .$$

We now choose c_1 so that for large r,

$$\frac{2\pi c}{\log r} \exp\left(\frac{2\pi c}{\log r}\right) \leq \frac{c_1}{\log r} . \qquad //$$

Let $t: S^3_r \to \mathbb{R}$ be the distance to p along S^3_r. Let $\theta = (\theta^1, \theta^2)$ be local coordinates on S^2_r, extended to S^3_r by requiring θ to be constant along each c_x.

Lemma 5.2. For any $x \in S^3_r$,

$$\left| \int_{c_x} T^{-1} F T \right| \leq \frac{2c}{r \log r} .$$

Proof. We suppose that x is on S^2_r, that c_x is taken to be the great arc from p to x, and that $d\theta^1$, $d\theta^2$ are orthonormal at x. The general case will follow easily. Note that

$$|d\theta^i| = \csc\left(\frac{t}{r}\right) , \quad |dt| = 1 .$$

Let

$$F = F_{01}\, dt \wedge d\theta^1 + F_{02}\, dt \wedge d\theta^2 + F_{12}\, d\theta^1 \wedge d\theta^2 ,$$

so

$$\int_{c_x} T^{-1}FT = \sum_{i=1}^{2} \left(\iint_{c_x} T^{-1}F_{0i}T \, dt \right) d\theta^i .$$

Thus

$$\left| \int_{c_x} T^{-1}FT \right| \leq \sum_{i=1}^{2} \int_{c_x} |F_{0i}| dt$$

$$\leq 2 \int_0^{\pi r/2} |F| \sin\left(\frac{t}{r}\right) dt$$

$$\leq 2 \cdot \frac{c}{r^2 \log r} \cdot r . \qquad //$$

If x is a point in the northern hemisphere (i.e. $t < \pi r/2$), then we have by Theorem 3.1,

$$A(x) = -dT \cdot T^{-1} + T\left(\iint_{c_x} T^{-1}FT \right) T^{-1} . \qquad (5.3)$$

If T_0 is constant, then substituting TT_0 for T leaves (5.3) unchanged. It follows that (5.3) is also valid if x is in the southern hemisphere. Hence we have

<u>Corollary 5.4</u>. (i) $|A + dT \cdot T^{-1}| \leq \frac{2c}{r \log r}$.

(ii) $|dT| \leq c_2 \frac{\log \log r}{r}$ for some constant c_2.

Given $\delta \in (0,1)$, let R be the set of points in S^3_r with $\frac{\pi r}{2} - \delta < t < \frac{\pi r}{2} + \delta$, and let

$$U(\theta) = \lim_{t \to \frac{\pi r}{2}-} T(t,\theta) .$$

From Corollary 5.4(ii), it follows that there exists a constant c_3 such that the inequality

$$|dU| \leq c_3 \frac{\log\log r}{r} \tag{5.5}$$

holds on R. By Lemma 5.1, we can choose δ sufficiently small that on R,

$$|U-T| \leq \frac{2c_1}{\log r} .$$

We suppose that $2c_1/\log r < \frac{1}{2}$ so that $f: R \to u(N)$ may be defined by the power series

$$f = \log U^{-1}T = -\sum_{n=1}^{\infty} \frac{1}{n}(1-U^{-1}T)^n .$$

Then

$$\begin{aligned} |f| &\leq \sum_{n=1}^{\infty} \frac{1}{n}|1-U^{-1}T|^n \\ &\leq \sum_{n=1}^{\infty} (\frac{2c_1}{\log r})^n \\ &\leq \frac{4c_1}{\log r} . \end{aligned} \tag{5.6}$$

Let $\phi: \mathbb{R} \longrightarrow [0,1]$ be a smooth function satisfying

$$\phi(t) = \begin{cases} 1 & |t| \geq 1 \\ 0 & |t| \leq \frac{1}{2} . \end{cases}$$

Define $\tilde{T}: R \to U(N)$ by

$$\tilde{T}(t,\theta) = U(\theta)\exp[\phi(\frac{t-\frac{\pi r}{2}}{\delta})f(t,\theta)]$$

and extend $\tilde{T}$ to be a smooth function on S^3_r by setting it equal to T on $S^3_r - R$.

Lemma 5.7. Let V and E be matrix valued functions.

(i) If $V = e^E$ then $|dV| \le e^{|E|}|dE|$

(ii) If $|1-V| \le \mu < 1$ and $E = \log V$, then

$$|dE| \le \frac{1}{1-\mu}|dV| .$$

Proof. This follows from differentiating the power series for exp and log. //

Lemma 5.8. There exist constants c_4, c_6, c_7 such that on R,

(i) $|df| \le c_4 \frac{\log\log r}{r}$

(ii) $\left|\frac{\partial\tilde{T}}{\partial t}dt\right| \le |d\tilde{T}| \le \frac{c_6}{\delta \log r}$

(iii) $\left|\sum_{i=1}^{2} \frac{\partial\tilde{T}}{\partial\theta^i} d\theta^i\right| \le c_7 \frac{\log\log r}{r}$

Proof. (i) Apply Lemma 5.7(ii) with $\mu = \frac{1}{2}$.

(ii) By Lemma 5.7(i),

$$|d\tilde{T}| \le |dU|e^{|f|} + e^{|f|}|df| + e^{|f|}\left|d[\phi(\frac{t-\frac{\pi r}{2}}{\delta})]\right| \cdot |f| .$$

Then, using (5.5), (5.6), and Lemma 5.8(i),

$$|d\tilde{T}| \le e^{4c_1/\log r}\left(c_3\frac{\log\log r}{r} + c_4\frac{\log\log r}{r} + \frac{c_5}{\delta}\cdot\frac{4c_1}{\log r}\right)$$
$$\le \frac{c_6}{\delta \log r} .$$

(iii) Similar, except that the term involving δ is absent. //

Lemma 5.9. For some c_9,

$$\left|\int_{S_r^3} \mathrm{tr}(T^{-1}dT)^3 - \int_{S_r^3} \mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3\right| \le c_9 \frac{(\log\log r)^2}{\log r} .$$

Proof.

$$\left|\int_{S^3_r} \mathrm{tr}(T^{-1}dT)^3 - \int_{S^3_r} \mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3\right| = \left|\int_R \mathrm{tr}(T^{-1}dT)^3 - \int_R \mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3\right|$$
$$\leq N\int_R |dT|^3 + \int_R |\mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3| \ .$$

By Lemma 5.8(ii) and 5.8(iii), there exists c_8 such that

$$|\mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3| \leq c_8 \cdot \frac{1}{\delta \log r} \cdot \left(\frac{\log\log r}{r}\right)^2 \ .$$

The volume of R is bounded by $4\pi r^2\delta$, so from Corollary 5.4(ii) and the above,

$$\left|\int_{S^3_r} \mathrm{tr}(T^{-1}dT)^3 - \int_{S^3_r} \mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3\right|$$
$$\leq N\cdot 4\pi r^2\delta\cdot\left(c_2\frac{\log\log r}{r}\right)^3 + 4\pi r^2\delta\cdot c_8\cdot\frac{1}{\delta\log r}\cdot\left(\frac{\log\log r}{r}\right)^2$$
$$\leq c_9\,\frac{(\log\log r)^2}{\log r} \ . \qquad //$$

We now complete the proof of Theorem 1.1. From $\mathrm{tr}\,F^2 = d\,\mathrm{tr}(AF - \frac{1}{3}A^3)$ and

$$|F| \leq \frac{c}{r^2\log r} \ ,$$

it follows that

$$\left|\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \mathrm{tr}\,F^2 - \frac{1}{8\pi^2}\int_{S^3_r} \mathrm{tr}(AF - \tfrac{1}{3}A^3)\right| \leq \frac{N}{8\pi^2}\int_r^\infty \left(\frac{c}{\rho^2\log\rho}\right)^2 2\pi\rho^3 d\rho = \frac{c_{10}}{\log r} \ .$$

From Proposition 4.2,

$$\left|\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \mathrm{tr}\,F^2 + \frac{1}{24\pi^2}\int_{S^3_r} \mathrm{tr}\,A^3\right| \leq c_{11}\,\frac{\log\log r}{\log r} \ .$$

Corollary 5.4(i) shows that the hypothesis of Proposition 4.3 is satisfied if $B = -dT \cdot T^{-1}$, so

$$\left| \frac{1}{8\pi^2} \int_{\mathbb{R}^4} \operatorname{tr} F^2 - \frac{1}{24\pi^2} \int_{S^3_r} \operatorname{tr}(T^{-1} dT)^3 \right| \leq c_{12} \frac{(\log \log r)^2}{\log r} .$$

By Lemma 5.9,

$$\left| \frac{1}{8\pi^2} \int_{\mathbb{R}^4} \operatorname{tr} F^2 - n \right| \leq c_{13} \frac{(\log \log r)^2}{\log r} ,$$

where n is the integer

$$\frac{1}{24\pi^2} \int_{S^3_r} \operatorname{tr}(\tilde{T}^{-1} d\tilde{T})^3 .$$

The proof of Theorem 1.1 is now finished by letting r tend to infinity.

6. An Example

Let $T: S^3 \to SU(2)$ be the standard identification. Let ω be the pull-back to $\mathbb{R}^4$ of $T^{-1}dT$ by the radial projection. The structural equation is

$$d\omega + \omega^2 = 0 .$$

Let $f: [0,\infty) \to \mathbb{R}$ be a smooth function satisfying

$$f(r) = \begin{cases} 0 & 0 \leq r \leq \frac{1}{2} \\ a & r \geq 1 \end{cases}$$

for some real number a. Using r for the radial coordinate, $f(r)\omega$ is a well-defined smooth form on $\mathbb{R}^4$, and we define it to be a connection on the trivial $SU(2)$ bundle.

We compute

$$F = f'dr \wedge \omega + (f^2 - f)\omega^2$$

so $F \in O(1/r^2)$. Also

$$F^2 = 2(f^2 - f)f'dr \wedge \omega^3$$

has compact support and

$$\begin{aligned} \frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 &= \frac{1}{8\pi^2}\cdot\int_0^\infty 2f'(r)(f^2(r)-f(r))dr\cdot\int_{S^3} \operatorname{Tr}\, \omega^3 \\ &= \frac{1}{8\pi^2}\cdot\left[\tfrac{2}{3}f^3(r) - f^2(r)\right]_0^\infty \cdot 12 \text{ volume } S^3 \\ &= 2a^3 - 3a^2 . \end{aligned}$$

By varying a, we may obtain any real number.

7. A Generalization

Theorem 1.1 generalizes to the following situation. Let G be a compact Lie group and let $\rho: G \to U(N)$ be a representation. Let F be the curvature of some connection on some principal G-bundle over $\mathbb{R}^{2n}$. Let

$$C_n = \frac{(-1)^{n+1}}{n} \operatorname{Tr}\left(\frac{\rho(F)}{2\pi i}\right)^n .$$

If the connection extends to a connection on some bundle over S^{2n}, then the Chern number [7] of the associated vector bundle is $\int_{\mathbb{R}^{2n}} C_n$. This is an integer and is always divisible by $(n-1)!$ (see [4, p. 77] or [5, p. 156]).

Theorem 7.1. Suppose $n > 1$. If there exists a constant C such that

$$|F| \leq \frac{C}{r^2 \log r}, \quad r \geq 2 ,$$

then

$$\frac{1}{(n-1)!} \int_{\mathbb{R}^{2n}} C_n$$

is an integer. Any integer value is possible if $G = SU(N)$ and $N \geq n$.

Proof. The proof requires only minor modifications of the proof of Theorem 1.1, which we now discuss. According to the Chern-Weil theory [4, p. 114],

$$\operatorname{Tr} F^n = nd\int_0^1 \operatorname{Tr}\{A[tF + (t^2-t)A^2]^{n-1}\}dt .$$

It follows as before that in the radial gauge,

$$\int_{\mathbb{R}^{2n}} \operatorname{Tr} F^n = n\int_0^1 (t^2-t)^{n-1}dt \cdot \lim_{r\to\infty} \int_{S_r^{2n-1}} \operatorname{Tr} A^{2n-1} .$$

An elementary integration by parts gives

$$\int_0^1 t^{n-1}(1-t)^{n-1}dt = \frac{[(n-1)!]^2}{(2n-1)!} .$$

Approximating A by $-d\tilde{T}\cdot\tilde{T}^{-1}$ for some smooth $\tilde{T}: S^3_r \to U(N)$, we find that

$$\frac{1}{(n-1)!}\int_{\mathbb{R}^{2n}} C_n$$

is approximated by

$$\frac{-1}{(2\pi i)^n}\cdot\frac{(n-1)!}{(2n-1)!}\int_{S^{2n-1}_r} \mathrm{Tr}(\tilde{T}^{-1}d\tilde{T})^{2n-1} .$$

This is an integer because it is the integral that gives Bott periodicity [3],

$$\pi_{2n-1}U(N) \cong \mathbb{Z} ,$$

for $n \leq N$. The integral is zero if $n > N$. //

The proof of Theorem 7.1 breaks down if $n = 1$. In order to estimate the holonomy around a closed curve near infinity, we expressed the curve as the boundary of a surface where the curvature was small and used Corollary 3.4. But the sphere of radius r in $\mathbb{R}^{2n}$ is simply-connected only if $n > 1$.

The following theorem shows that Theorem 7.1 actually fails if $n = 1$.

Theorem 7.2. Let F be a two-form on $\mathbb{R}^2$ with values in some Lie algebra. Then F is the curvature of some connection on a principal bundle over $\mathbb{R}^2$.

Proof. Write

$$F = f(x,y)\, dx \wedge dy .$$

Then F is the curvature of the connection

$$A(x,y) = \left(\int_0^x f(x',y)dx'\right)dy \ . \qquad //$$

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